

Zion Leonahenahe Basque

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Arizona State University

Education

- 2021–2026 **Ph.D. in Computer Science**
Arizona State University
Advisors: Ruoyu "Fish" Wang and Yan Shoshitaishvili
Thesis: Toward a Science of Software Reverse Engineering
- 2024 **M.S. in Computer Science**
Arizona State University
- 2017–2020 **B.S. in Computer Science**
Arizona State University

Publications

Peer-reviewed Proceedings

- [C1] **Zion Leonahenahe Basque**, Samuele Doria, Ananta Soneji, Wil Gibbs, Adam Doupé, Yan Shoshitaishvili, Eleonora Losiouk, Ruoyu Wang, and Simone Aonzo, "Decompiling the Synergy: An Empirical Study of Human–LLM Teaming in Software Reverse Engineering," in *Proceedings of the 2026 Network and Distributed System Security Symposium*, 2026.
- [C2] Zehua Zhang, Ati Priya Bajaj, Divij Handa, Siyu Liu, Arvind S Raj, Hongkai Chen, Hulin Wang, Yibo Liu, **Zion Leonahenahe Basque**, Souradip Nath, Vishal Juneja, Nikhil Chapre, Yan Shoshitaishvili, Adam Doupé, Chitta Baral, and Ruoyu Wang, "BuildBench: Benchmarking LLM Agents on Compiling Real-World Open-Source Software," in *NeurIPS 4th Deep Learning for Code Workshop*, 2025.
- [C3] Irina Ford, Ananta Soneji, Faris Bugra Kokulu, Jayakrishna Vadayath, **Zion Leonahenahe Basque**, Gaurav Vipat, Adam Doupé, Ruoyu Wang, Gail-Joon Ahn, Tiffany Bao, *et al.*, "'Watching over the shoulder of a professional': Why Hackers Make Mistakes and How They Fix Them," in *Proceedings of the 45th IEEE Symposium on Security and Privacy*, 2024. (17.8% acceptance rate.)
- [C4] Wil Gibbs, Arvind S Raj, Jayakrishna Menon Vadayath, Hui Jun Tay, Justin Miller, Akshay Ajayan, **Zion Leonahenahe Basque**, Audrey Dutcher, Fangzhou Dong, Xavier Maso, *et al.*, "Operation Mango: Scalable discovery of Taint-Style vulnerabilities in binary firmware services," in *Proceedings of the 33rd USENIX Security Symposium*, 2024. (18.32% acceptance rate.)

- [C5] Kuntal Kumar Pal, Ati Priya Bajaj, Pratyay Banerjee, Audrey Dutcher, Mutsumi Nakamura, **Zion Leonahenahe Basque**, Himanshu Gupta, Saurabh Arjun Sawant, Ujjwala Anantheswaran, Yan Shoshitaishvili, Adam Doupé, Chitta Baral, and Ruoyu Wang, “Len or index or count, anything but v1’: Predicting Variable Names in Decompilation Output with Transfer Learning,” in *Proceedings of the 45th IEEE Symposium on Security and Privacy*, 2024. (17.8% acceptance rate.)
- [C6] **Zion Leonahenahe Basque**, Ati Priya Bajaj, Wil Gibbs, Jude O’Kain, Derron Miao, Tiffany Bao, Adam Doupé, Yan Shoshitaishvili, and Ruoyu Wang, “Ahoy SAILR! There is No Need to DREAM of C: A Compiler-Aware Structuring Algorithm for Binary Decompilation,” in *Proceedings of the 33rd USENIX Security Symposium*, 2024. (18.32% acceptance rate.)
- [C7] Hui Jun Tay, Kyle Zeng, Jayakrishna Menon Vadayath, Arvind S Raj, Audrey Dutcher, Tejesh Reddy, Wil Gibbs, **Zion Leonahenahe Basque**, Fangzhou Dong, Zack Smith, Adam Doupé, Tiffany Bao, Yan Shoshitaishvili, and Ruoyu Wang, “Greenhouse: Single-Service Relocating of Linux-Based Firmware Binaries in User-Space Emulation,” in *Proceedings of the 32nd USENIX Security Symposium*, 2023. (29.2% acceptance rate.)
- [C8] Penghui Zhang, Zhibo Sun, Sukwha Kyung, Hans Behrens, **Zion Leonahenahe Basque**, Haehyun Cho, Adam Oest, Ruoyu Wang, Tiffany Bao, Yan Shoshitaishvili, Gail-Joon Ahn, and Adam Doupé, “I’m Spartacus, No, I’m Spartacus: Proactively Protecting Users from Phishing by Intentionally Triggering Cloaking Behavior,” in *Proceedings of the 2022 ACM Conference on Computer and Communications Security*, 2022. (22.4% acceptance rate.)

Preprints

- [R1] Muqi Zou, Hongyu Cai, Hongwei Wu, **Zion Leonahenahe Basque**, Arslan Khan, Berkay Celik, Dave Tian, Antonio Bianchi, Ruoyu Wang, and Dongyan Xu, *D-LiFT: Improving LLM-based Decompiler Backend via Code Quality-driven Fine-tuning*, (In submission), 2025.

Academic Service

Conference & Workshop Organization

2026	Artifact Evaluation Committee Chair , USENIX WOOT Conference on Offensive Technologies
2025	Founding Program Chair , ACM CCS Workshop on Software Understanding and Reverse Engineering (SURE)
2025	CTF Committee Chair , Annual Computer Security Applications Conference (ACSAC)
2024	CTF Committee , Annual Computer Security Applications Conference (ACSAC)

Program Committee

2026	Workshop on Datasets for Software Security (Data4SoftSec)
2026	AAAI Symposium on Human-Aware AI Agents for the Cyber Battlefield
2026	European Workshop on Systems Security (EuroSec)

2026	Conference on Detection of Intrusions and Malware & Vulnerability Assessment (DIMVA)
2026	USENIX WOOT Conference on Offensive Technologies
2025	Cyber Security Experimentation and Test Workshop (CSET)
2025	USENIX WOOT Conference on Offensive Technologies

Reviewer

2025	ACM Transactions on Privacy and Security (TOPS)
2025	Expert Systems with Applications Journal

Invited Talks & Presentations

Invited Talks

- [T1] *ARTIPHISHELL Intelligence and LLM Autonomy in Patching*. **Microsoft Security**, 2025.
- [T2] *Buccaneers of the Binary: Plundering Compiler Optimizations for Decompilation Treasure*. **Reverse Conference**, 2025.
- [T3] *Cyber Reasoning Systems in the Age of LLMs*. **University of Texas: El Paso**, 2025.
- [T4] *Falling in Love with CTFs: Lessons from Shellphish in Love and Loss*. **United States Cyber Games, Keynote**, 2025.
- [T5] *ARTIPHISHELL Intelligence: Advancing Autonomous Software Security with AI*. **Oracle**, 2024.
- [T6] *Improving the Understanding of Software Through Human-Centric Decompilation*. **University of Hawaii, Manoa**, 2024.
- [T7] *Toward Human-Centric Decompilation*. **NSA Research**, 2024.
- [T8] *Bridging the gap in the static and dynamic analysis of binaries through decompiler tomfoolery*. **CactusCon**, 2023.
- [T9] *Modern Approaches in Human-Centric Decompilation*. **Ohio State University**, 2023.
- [T10] *Your Teammate Isn't Human! Mixing Decompilation and AI for Modern Reverse Engineering*. **HITCON, Taiwan, Keynote**, 2023.

Panels

- [L1] *Beyond decompilation: multi-level lifting for automatic software understanding*. **REcon, Québec, Canada, Panelist**, 2025.

Posters

- [P1] *Goal-oriented binary decompilation is key to understanding complex binary programs*. **Software for National Security Partnership Forum**, 2025.

Demonstrations

- [D1] *BinSync: automatic lifting of cross-analyses binary data*. **USENIX WOOT Conference on Offensive Technologies**, 2024.

Teaching

Arizona State University

- 2025 **Teaching Assistant**, Applied Program Analysis and Debugging (CSE 598); 16 students. Partial lecturing and grading assignments.
- 2025 **Instructor**, Cybersecurity History and Culture (CSE 194); 31 students. Designed syllabus; authored new lectures and assignments; coordinated two external invited speakers.
- 2021 **Teaching Assistant & Platform Developer**, Systems Security (CSE 466); 106 students. Developed two pwn.college modules used across dozens of U.S. universities; supported instruction.
- 2020 **Teaching Assistant**, Software Security (CSE 545); 112 students. Developed new assignments; supported instruction and course operations.

Mentorship

Undergraduate

- 2025 Justin Yen, Arizona State University
BinSync server improvements for use in government and commercial decompilers
- 2025 Nishanth Pallapu, Arizona State University
Automatic and assisted merging of conflicts in collaborative reverse engineering
- 2025 Luke Perez, Arizona State University
Visualizing reverse engineering progress at scale
- 2024 Caden Sounart, Arizona State University
Investigating flawed cryptography protocol implementations

High School

- 2023 Saketh Reddy, Now: **Georgia Tech**, AIxCC 1st-place contributor
Automatic detection of malicious extensions in audio systems
- 2023 Derron Miao, **USENIX Security 2024 publication**
Simplifying decompiler output through control flow linearization

Honors & Awards

- 2024 **Defense Advanced Research Projects Agency (DARPA)**. AI Cyber Challenge Finalist (\$2 million award).
- 2022 **National Science Foundation**. Graduate Research Fellowship, Honorable Mention.
- 2021 **Arizona State University**. Dean's Fellowship.
- 2021 **Computing Research Association**. Undergraduate Research Award, Honorable Mention.
- 2020 **Arizona State University**. Graduate Impact Award.
- 2017–2023 **Private & Public Scholarships**.

Center for Cyber Safety and Education, APIA/Coca-Cola Foundation, Pauahi Foundation, Hawaii Community Foundation THINK, Hokuli'a Foundation, David and Julia Desha Foundation. (\$171k in awards.)

Selected Media Coverage

2024	Fox 10 News , "ASU students win \$2M in cybersecurity competition"
2024	The Washington Post , "Hackers race to win millions in contest to thwart cyberattacks with AI"
2021	Computing Research Association , "From CTF to PhD: Hacker turned Security Researcher"
2021	Arizona State University , "Cybersecurity competition challenges next generation of security experts"
2021	Amazon Prime Video , "The College Tour Season 1: Arizona State University"

Industry Experience

2024-	<i>Founder</i> , Ka Moku Technologies Founded to transition research prototypes in binary analysis and reverse engineering into industry solutions. Primary work through contracting.
2019	<i>Fuzzing Research Intern</i> , Mayhem Security Developed fuzzing synchronization systems to discover vulnerabilities autonomously. Discovered two zero-days in Netflix and bootloader software.

Other Experience

- **Led ASU technical teams on DARPA programs** as technical lead for AMP and FIRE (2024); contributor to HACCS and CHESS.
- **Co-captain, Shellphish** (2020–2023), the oldest ethical hacking team in the U.S.; competed in 124+ 48-hour CTFs and organized two editions of iCTF with 200 teams worldwide.
- **Led autonomous vulnerability patching team** for Shellphish's DARPA AIXCC submission; placed 5th overall and built the competition's most accurate patching system.
- **Developed open-source security tools** adopted by the community. Creator of BinSync, DAILA, and decomp2dbg. Maintainer of the angr decompiler. Accumulated 3k+ GitHub stars; hundreds of installs per day.
- **Contributor of widely used education platform**, pwn.college, which has been adopted by organizations such as the Department of Defense, University of Hawaii, and other institutions across the globe.